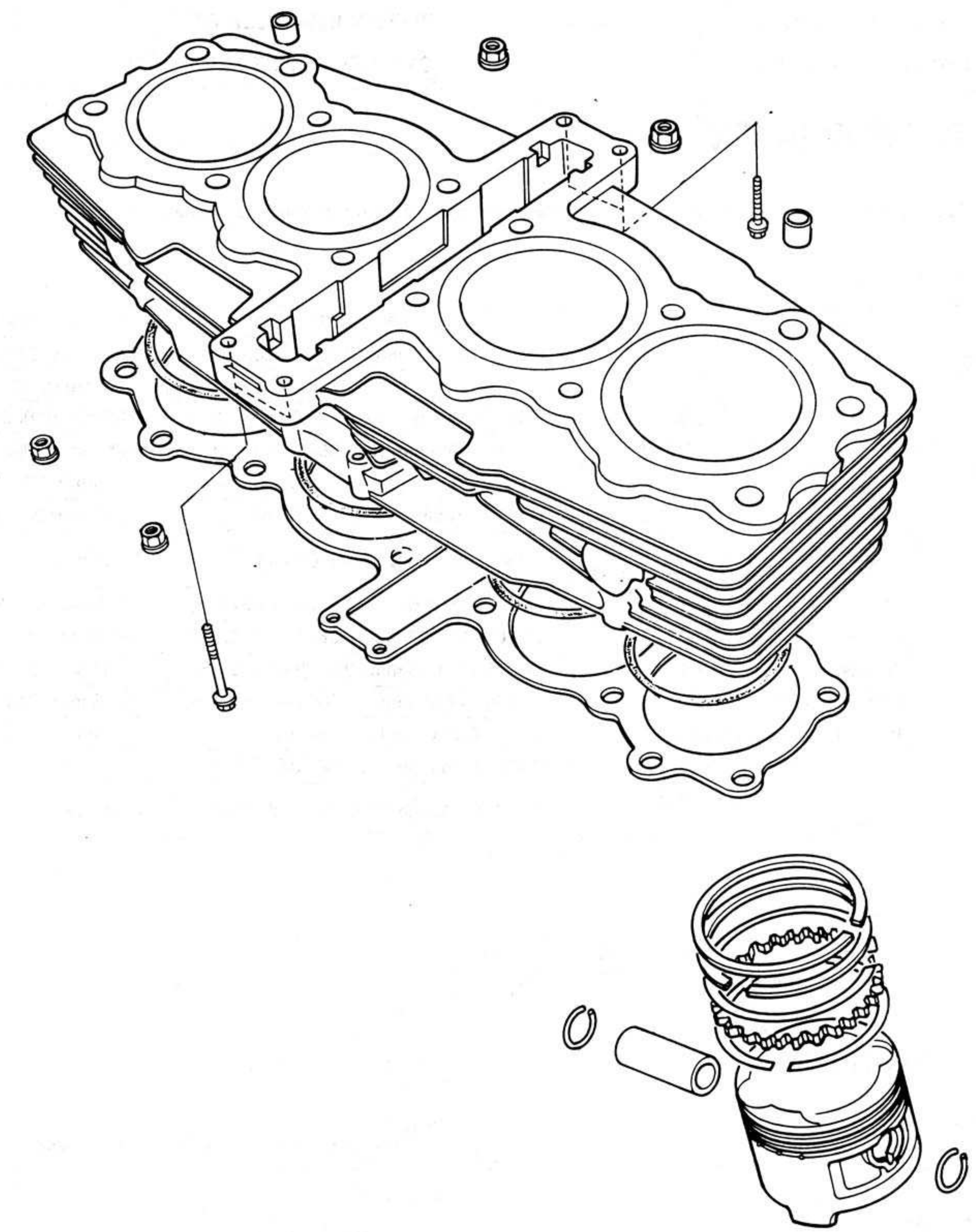


SERVICE INFORMATION

GENERAL

All cylinder/piston maintenance and inspection can be accomplished without removing the engine from the frame.

SPECIFICATIONS

ITEM			STANDARD	SERVICE LIMIT
Cylinder	I.D.		67.000-67.010 mm (2.6378-2.6382 in)	67.10 mm (2.642 in)
Piston, Piston rings and piston pin	Warpage			0.10 mm (0.004 in)
	Piston ring to ring groove clearance	TOP	0.015-0.045 mm (0.0006-0.0018 in)	0.06 mm (0.002 in)
		SECOND	0.015-0.045 mm (0.0006-0.0018 in)	0.06 mm (0.002 in)
	Ring end gap	TOP	0.15-0.30 mm (0.006-0.012 in)	0.5 mm (0.02 in)
		SECOND	0.15-0.30 mm (0.006-0.012 in)	0.5 mm (0.02 in)
		OIL (SIDE RAIL)	0.30-0.90 mm (0.012-0.035 in)	1.1 mm (0.04 in)
	Piston O.D.		66.960-66.990 mm (2.6362-2.6374 in)	66.90 mm (2.634 in)
	Piston pin bore		17.002-17.008 mm (0.6694-0.6696 in)	17.05 mm (0.671 in)
	Connecting rod small end I.D.		17.016-17.034 mm (0.6699-0.6706 in)	17.07 mm (0.672 in)
	Piston pin O.D.		16.994-17.000 mm (0.6691-0.6693 in)	16.98 mm (0.669 in)
	Piston to piston pin clearance		0.002-0.014 mm (0.0001-0.0006 in)	0.04 mm (0.002 in)
	Cylinder to piston clearance		0.010-0.050 mm (0.0004-0.0020 in)	0.10 mm (0.004 in)
	Piston pin to connecting rod clearance		0.016-0.040 mm (0.0006-0.0016 in)	0.06 mm (0.002 in)

TOOLS

Special

Piston base (2 required)	07958-3000000
Piston ring compressor	07954-2830000

TROUBLE SHOOTING

Compression low
1. Worn cylinder or piston rings
2. Leaking valve seats

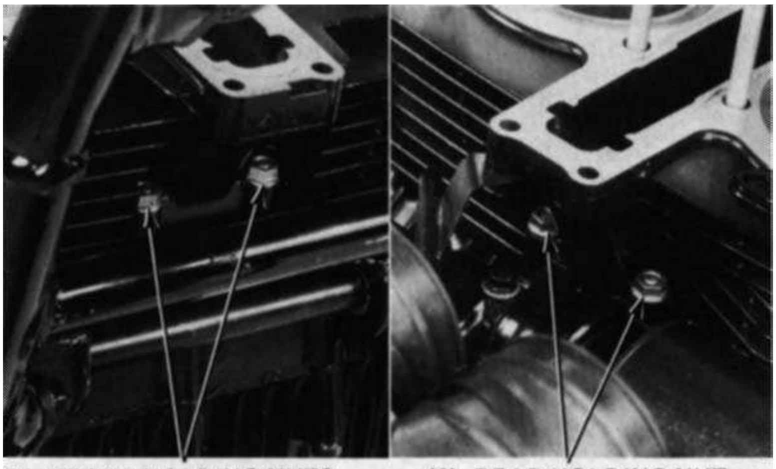
Excessive smoke
1. Worn cylinder or piston
2. Improper installation of piston rings
3. Scored or scratched piston or cylinder wall

Overheating
1. Excessive carbon build up on the piston or combustion chamber wall
2. Incorrect spark plug

Knocking or abnormal noise
1. Worn piston or cylinder
2. Excessive carbon build up
3. Low octane fuel

CYLINDER REMOVAL

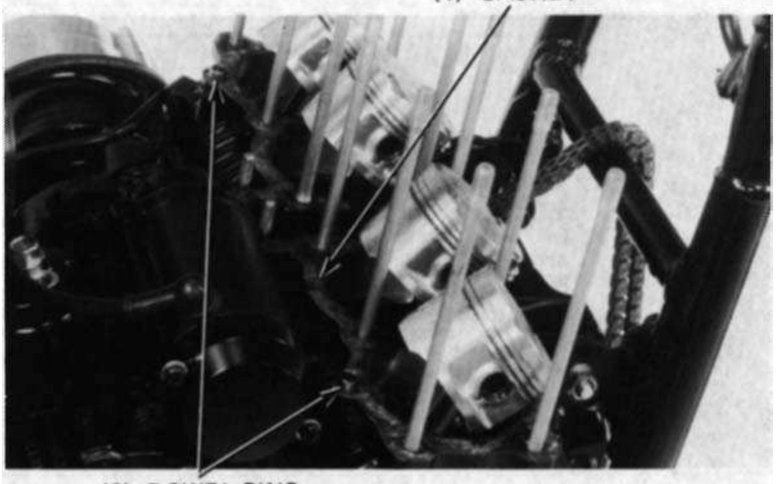
Remove the cylinder head (Section 6).
Remove the cam chain tensioner guide.
Remove the front and rear cylinder holding nuts and remove the cylinder.



(1) FRONT HOLDING NUTS (2) REAR HOLDING NUT

Remove the cylinder gasket and dowel pins.

(1) GASKET

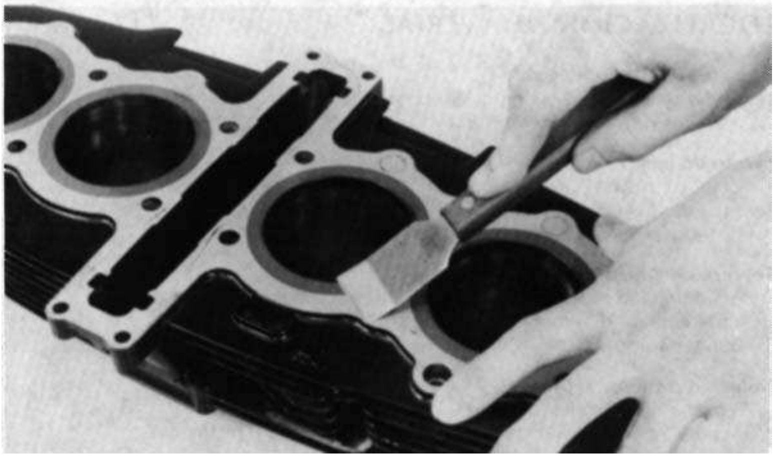


(2) DOWEL PINS

Clean off the cylinder gasket surfaces.

NOTE:
Gasket will come of easier if soaked in solvent.

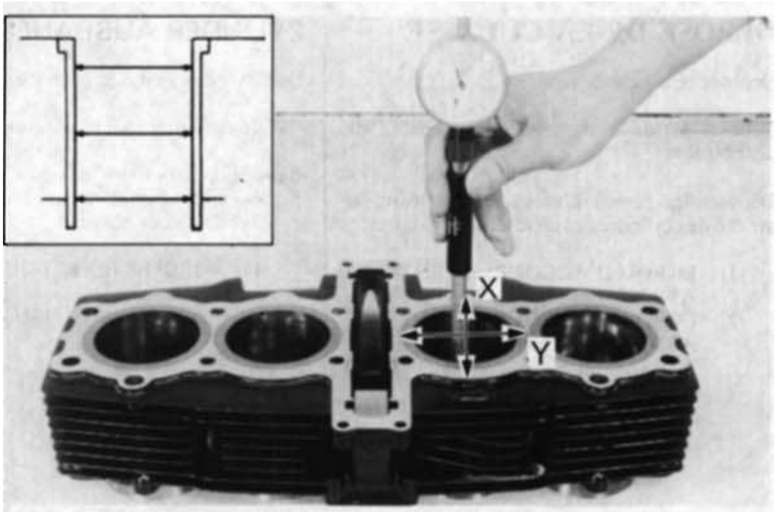
CAUTION
Do not damage the gasket surfaces.



CYLINDER INSPECTION

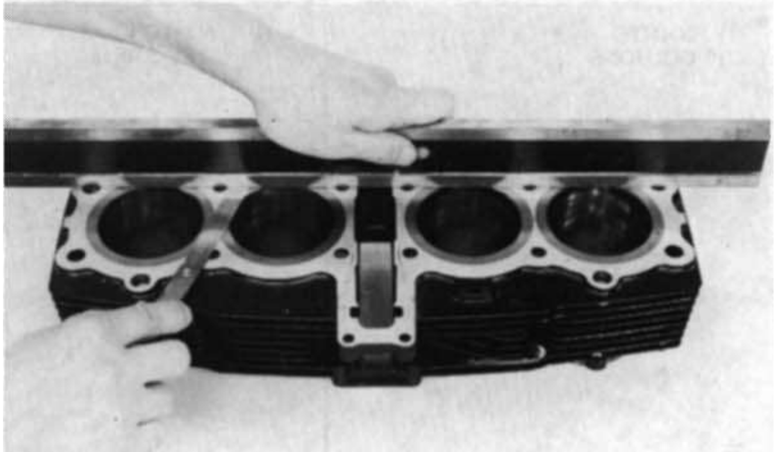
Inspect the cylinder bores for wear or damage.
Measure the cylinder I.D. at three levels in X and Y axis.

SERVICE LIMIT: 67.10 mm (2.642 in)



Inspect the top of the cylinder for warpage.
Check in an X pattern as shown.

SERVICE LIMIT: 0.10 mm (0.004 in)



PISTON REMOVAL

Place rags in the crankcase openings.
Remove each piston pin clip with needle nose pliers being careful not to allow clips to fall into the crankcase.
Press the piston pins out.
Mark each piston to indicate its cylinder position for reassembly.

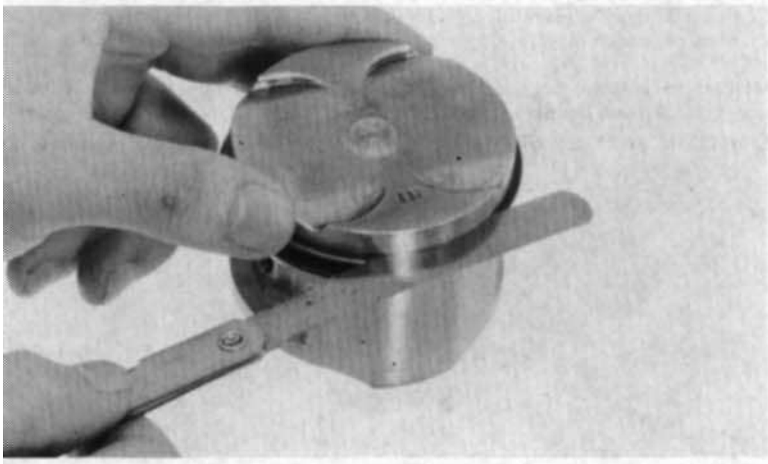
PISTON/PISTON RING INSPECTION

Inspect the piston ring-to-groove clearance.

SERVICE LIMIT:
TOP: 0.06 mm (0.002 in)
SECOND: 0.06 mm (0.002 in)

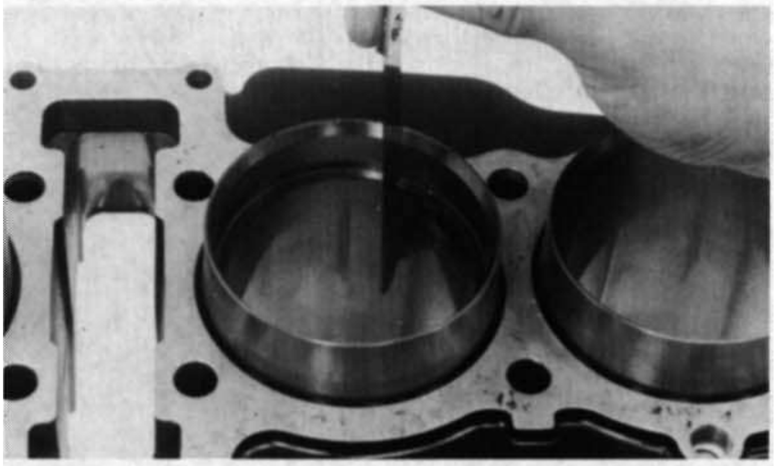
Mark the rings so that they can be returned to correct piston during reassembly.

Inspect the pistons for damage or cracks; ring grooves for wear.



Inspect each piston ring into the cylinder, and inspect the end gap.

SERVICE LIMITS:
TOP: 0.050 mm (0.020 in)
SECOND: 0.050 mm (0.020 in)
OIL (side rail) 1.10 mm (0.043 in)



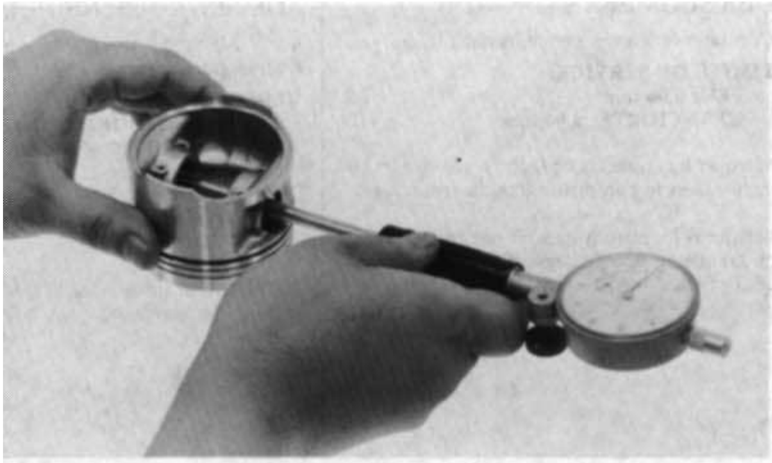
Measure the piston O.D. 14 mm (0.6 in) from the bottom of the skirt and 90 deg to the piston pin hole.

SERVICE LIMIT: 66.90 mm (2.634 in)

Calculate the cylinder-to-piston clearance.

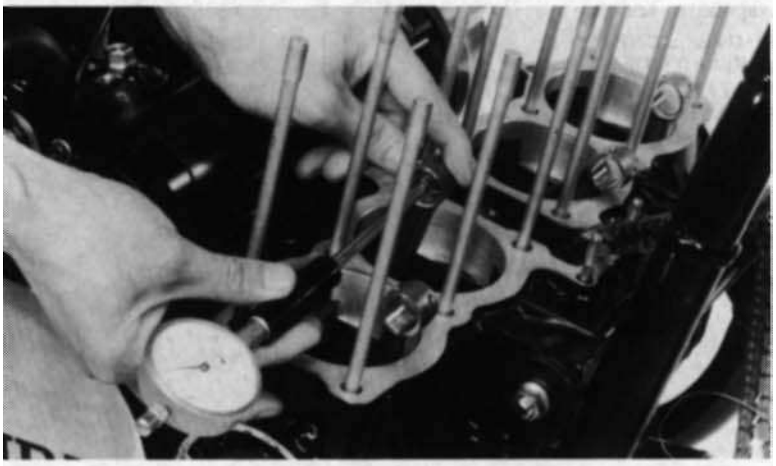
SERVICE LIMIT: 0.10 mm (0.004 in)

Measure the piston pin hole I.D.



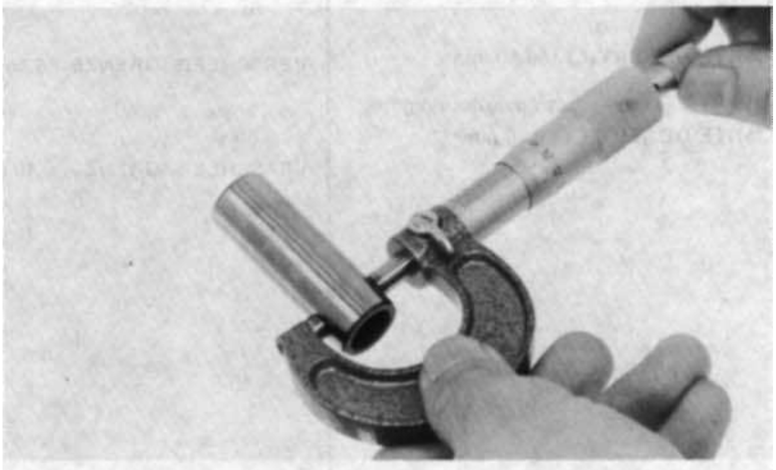
SERVICE LIMIT: 17.05 mm (0.671 in)

Measure the connecting rod small end I.D. (See Section 12 for replacement procedure.)



SERVICE LIMIT: 17.07 mm (0.672 in)

Measure the piston pin O.D.



SERVICE LIMIT: 16.98 mm (0,669 in)

Calculate the piston pin-to-piston clearance.

SERVICE LIMIT: 0.04 mm (0.002 in)

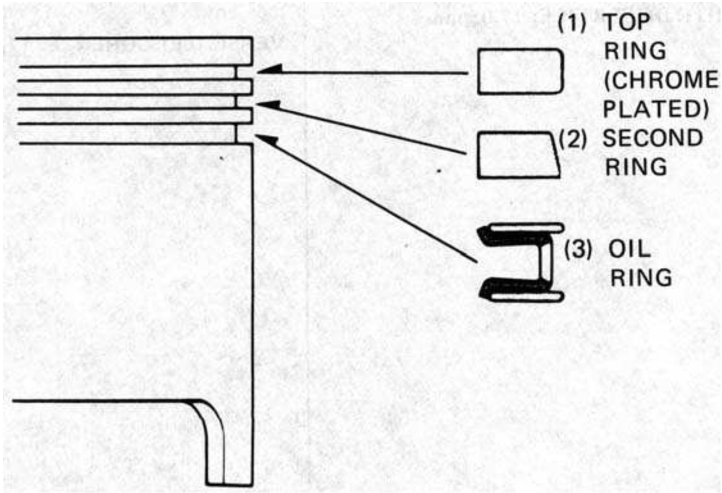
Calculate the piston pin-to-connecting rod clearance.

SERVICE LIMIT: 0.06 mm (0.002 in)

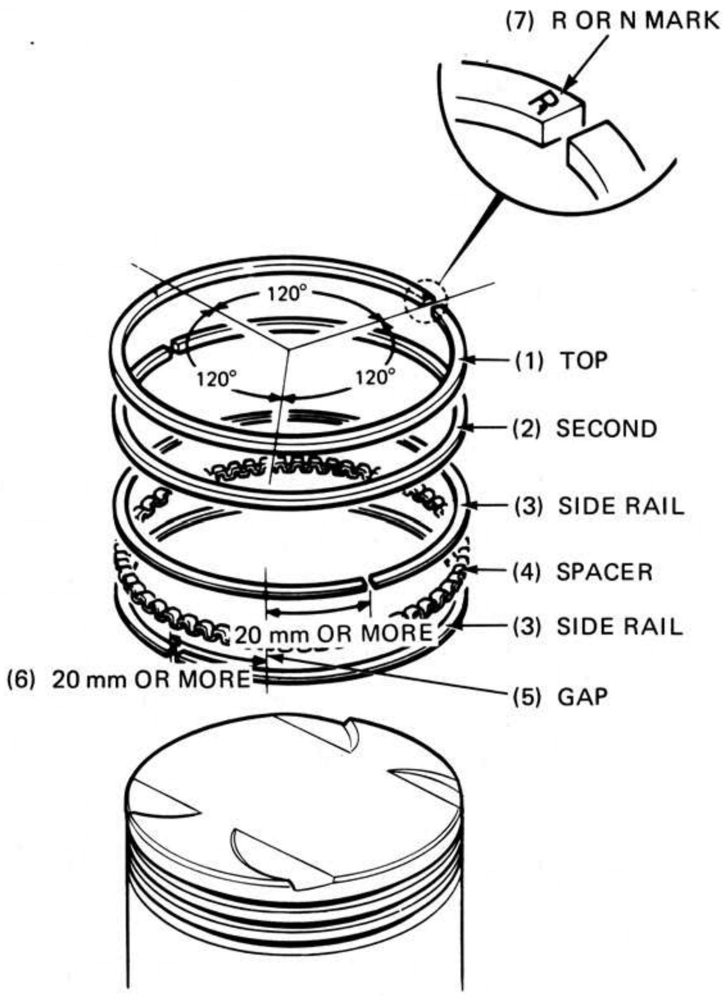
PISTON RING INSTALLATION

Install the piston rings with the markings face up

NOTE:
After installation, the rings should rotate freely in the grooves.

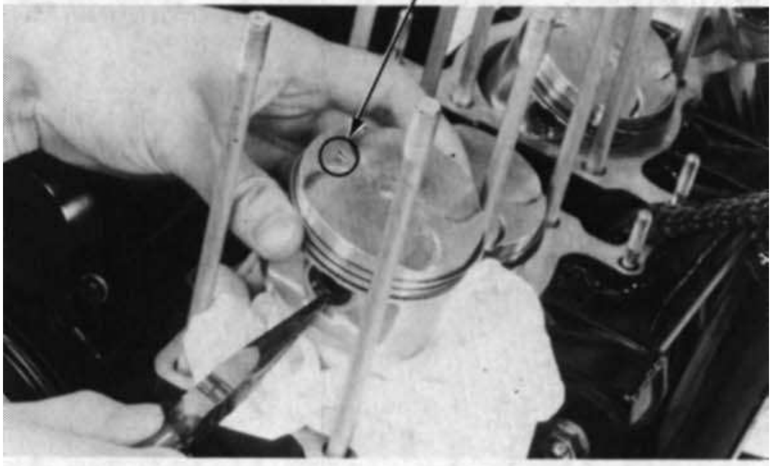


Space the piston ring end gaps 120 degrees apart.
Do not align the gaps in the oil rings.



PISTON INSTALLATION

Apply molybdenum disulphide grease to the connecting rod small ends.
Install the pistons, piston pins and clips. Be careful not to drop clips into the crank case,
NOTE:
- **Position the "IN" mark on the piston crown toward the intake side.**
- **Install the pistons in there original positions.**

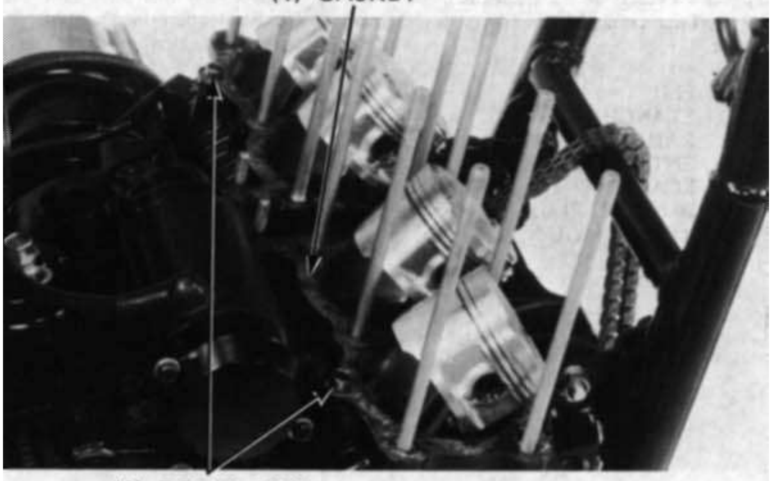


CYLINDER INSTALLATION

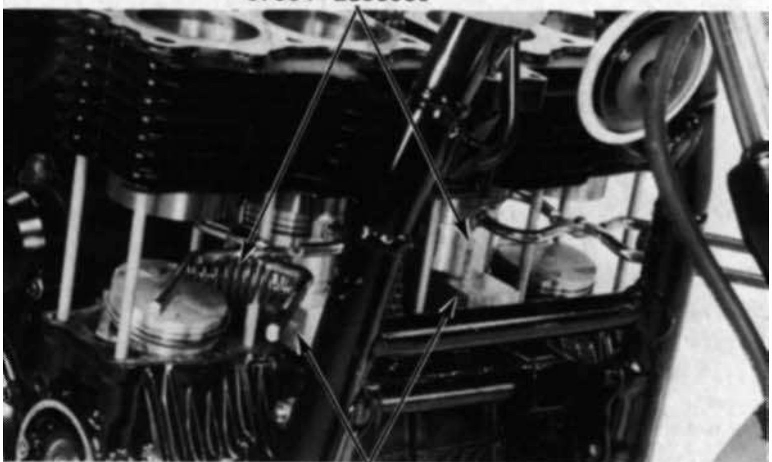
Clean the cylinder gasket surface of the crankcase.

CAUTION
Be careful not to damage the gasket surface.

Install the dowel pins and a new gasket.



Put the No. 2 and 3 pistons at TDC.
Place piston bases under the pistons, compress the piston rings with piston ring compressors and slide the cylinder over the No. 2 and 3 pistons.
Remove the piston ring compressors and piston bases.



Compress the No. 1 and 4 pistons with the piston ring compressors and slide the cylinder over the pistons.
Remove the piston ring compressors.

Install the cylinder holding nuts securely.

Install the cylinder head (Section 8).